RIS-Assisted mmWave/THz Communication Systems

Using a Smart Reflecting Surface to Overcome Signal Blockage in 6G Wireless Networks

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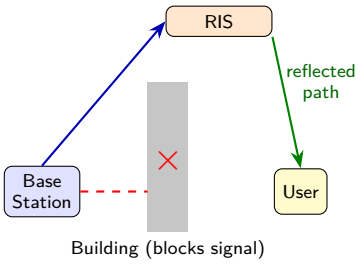
Minor Project – First Presentation

Outline

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Introduction: The Blockage Problem in 6G

- 6G networks will use **very high frequencies** (mmWave and THz) to deliver extremely fast data rates.
- Such signals behave almost like **light**: they travel in straight lines and cannot bend around objects.
- A building, a wall, or even trees can **completely block** the path between the tower (BS) and the user (UE).
- **Question:** how do we still reach a user who is hidden behind an obstacle?



Motivation: What is a Reconfigurable Intelligent Surface?

- An **RIS** is a thin panel made of many tiny reflecting elements, mounted on a wall or building.
- Each element can be **electronically tuned** to change how it reflects the signal – like a **smart mirror** that can be steered without moving.
- When all elements are tuned together, their reflections **add up** at the user and create a strong new path around the obstacle.

Advantages

- Passive: no power amplifiers, very low energy
- Cheap and easy to install on walls
- Bigger panel $\Rightarrow$ much stronger signal

Challenges

- The signal travels two hops, so it gets weaker
- Tower and RIS must be tuned **together**
- Finding the best tuning is a hard problem

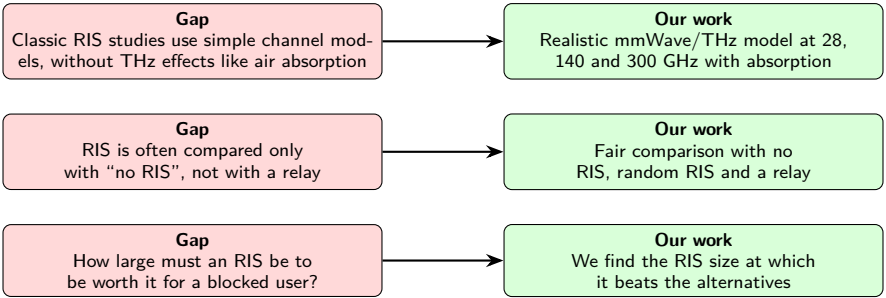
Objectives

1. Build a realistic simulation of a 6G link where the direct path is **blocked** and an RIS provides an alternative path.
2. **Maximize the data rate** delivered to the user(s) by jointly tuning the tower's beam and the RIS reflections.
3. Design an efficient algorithm – **Alternating Optimization (AO)** – to solve this tuning problem.
4. Compare our design with common alternatives: **no RIS**, an **untuned (random) RIS**, and a **relay** station.
5. Study how RIS size, transmit power, distance and frequency affect performance and **energy efficiency**.

Literature Survey

Paper	Key idea	Use in our work
Wu & Zhang, IEEE TWC 2019	Tune tower and RIS **alternately**; signal grows with RIS size	Basis of our algorithm
Di Renzo *et al.*, IEEE JSAC 2020	RIS vs. relays; RIS links suffer a two-hop loss	Relay comparison
Yu, Xu & Schober, IEEE ICCC 2019	Efficient RIS tuning with a multi-antenna tower	Multi-user tuning
Huang *et al.*, IEEE TWC 2019	Energy efficiency of RIS systems	Energy model
El Ayach *et al.*, IEEE TWC 2014	Realistic mmWave channel model (few strong paths)	Channel model
Huang, Mo & Yuen, IEEE JSAC 2020	Deep reinforcement learning (AI) for RIS	AI comparison

Research Gaps and Our Contribution



Proposed Methodology

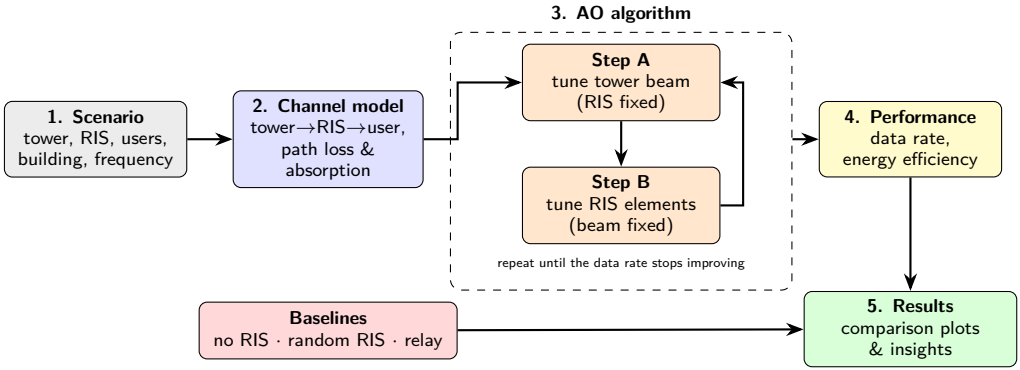
- **Two "knobs" to tune:** the tower's beam $\mathbf{W}$ and the reflection setting θ_m of every RIS element.
- **Goal:** choose them to give the user the highest data rate:

Main formula (from the problem statement)

$$\max_{\mathbf{W},\,\boldsymbol{\theta}} \; R = \log_2(1 + \mathrm{SINR}) \qquad \text{subject to} \quad \|\mathbf{W}\|^2 \le P_{\max}, \quad |\theta_m| = 1$$

- **SINR** = strength of the useful signal compared to noise and interference. Higher SINR $\Rightarrow$ more data per second.
- **Limits:** the tower has a power budget, and RIS elements can only *redirect* the signal, not amplify it.
- **Our approach – Alternating Optimization:** tuning both knobs at once is hard, so we **tune one at a time** and repeat until the data rate stops improving.

Workflow / System Architecture

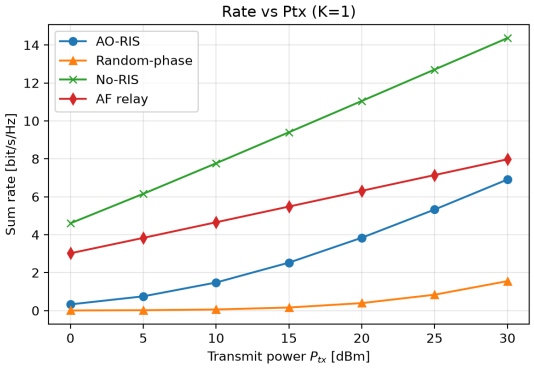


Implemented in **Python**; each result is averaged over **200 random channel conditions**.

Work Done So Far

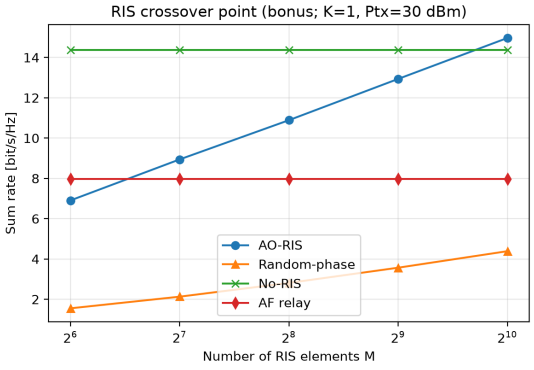
- **Done:** blocked-path scenario and realistic channel model at 28 / 140 / 300 GHz.
- **Done:** AO algorithm for one user and for four users.
- **Done:** three baselines – no RIS, random RIS, relay.
- **Done:** all four plots required by the problem statement, plus distance and frequency studies.
- **Done:** automated tests that check the results are physically correct.
- **Tried:** an AI method (reinforcement learning) as an alternative – it has not matched AO yet.
- **Next:** more realistic conditions (imperfect channel knowledge, hardware limits).
- **Next:** write the results as a research paper.

Result 1: Does Tuning the RIS Help?



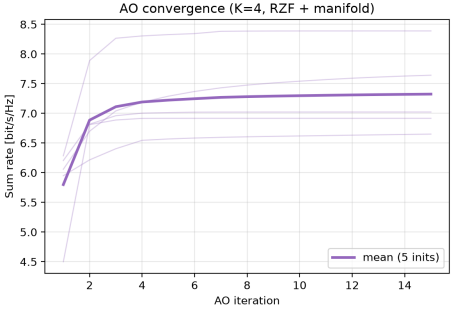
- **What it shows:** data rate as the tower's transmit power increases (higher = better).
- **Tuned RIS (blue)** gives about **4.4× more data** than an untuned RIS (orange).
- With a small RIS (64 elements), the relay and the weakened direct path are still better.
- **Why?** The reflected signal travels two hops and loses strength on both ⇒ the RIS must be large enough.

Result 2: How Big Should the RIS Be?

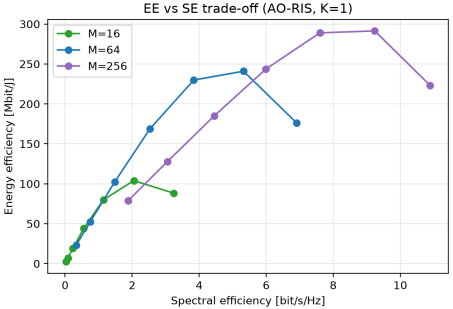


- **What it shows:** data rate as the number of RIS elements grows (64 $\rightarrow$ 1024).
- **Doubling the elements gives about 4$\times$ the signal power** – the tuned reflections add up together.
- The tuned RIS beats the **relay at $\sim$128 elements** and the **blocked direct path at 1024 elements**.
- At 140 GHz, 1024 elements fit on a panel of only about **3.4 cm $\times$ 3.4 cm**.

Result 3: Is the Algorithm Fast and Energy-Efficient?



- Each faint line is a different starting point; bold = average.
- Data rate **never drops** and settles within ∼**10 rounds** (4 users).



- Energy efficiency = data delivered per unit of energy.
- A **larger RIS** gives both **more data and better energy efficiency**; too much power wastes energy.

Conclusion and Future Work

Conclusion

- An RIS can **reconnect a blocked user** in 6G networks.
- **Tuning matters:** our AO algorithm delivers 2–4$\times$ more data than an untuned RIS.
- **Size matters:** the RIS must be large enough to overcome the two-hop loss.
- The algorithm is fast and always improves the data rate.

Future Work

- Imperfect channel knowledge
- Hardware limitations of the RIS
- Power-saving objective (minimum power)
- Better AI (reinforcement learning) method
- Research paper on the results

Thank You

Questions?

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